

A BRIEF INTRODUCTION TO APPLICATION PROGRAMMING INTERFACE (API)

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Abstract:

An API is a **contract** between pieces of applications serving the main software once integrated into the source code of the main application. These pieces of applications communicate with each other in the language they both understand and over a network if needed (Jacobson, et al., 2012). It grants the functionality of one application to another when called (Sommerville, 2016). This paper briefly explores the concept of Application Programming Interface (API) and covers a range of topics such as the security and the functionality of APIs and various API protocols and technologies that exist to fulfill the needs of both developing entities and end-users.

Concept:

An **application programming interface**, or **API**, enables commercial, military or private entities to make the data and functionality of their applications or systems available to external third-party developers, commercial partners, and internal departments within their own organizations. The usage of a defined interface enables services and products to interact with one another and benefit from each other's information and features. This interface is manipulated and programmed by developers to interact with other software, services or systems; they are not required to understand how a specific API is developed, and neither are the software's end-users. In short, an API is a **contract** between pieces of applications serving the main software once integrated into the source code of the main application. These pieces of applications communicate with each other in the language they both understand and over a network if needed (IBM Corporation, 2012; Jacobson, et al., 2012; Sommerville, 2016; Red Hat, Inc., 2022).

Functionality, Usability, and Drawbacks of APIs:

To expound and clarify this section clearly and accurately, let's first see the example of a bottle of cologne.

We apply the cologne that we purchase. Unless we are involved in the production process, medical professionals, health-conscious individuals, or looking into the manufacturer and their cologne manufacturing method because their product caused physical harm to their customers, we are not required to be aware of the cologne's manufacturing process. Due to improper production techniques, specific batches of cologne can become dangerous and life-threatening. Buyers are always satisfied when they use quality cologne.

Similarly, unless we are the developers or stakeholders involved in the development process of a computer application software or program, we necessarily do not have to know how various kinds of Application Programming Interfaces or APIs from different sources have been utilized to serve the specific-purpose application software or program we are using as an end-user. However, lousy development procedures from the development team's end can result in unwanted hindrances and drawbacks, such as the application not working, personal data being breached by third-party Blackhat hackers or individuals, and so on ([IBM Corporation, 2012](#)).

Consider a scenario where a group of friends from Vienna, Austria, travelled to Bucharest, Romania. However, they should have hired a tour guide who could show them around beforehand, and none was available when they remembered it. It turns out they have network connectivity and access to their devices. They may each download the Bucharest Touring Map app from their respective smartphone's app store. They do not need to be aware of the design process.

It is important to note, however, that the creators of that application used a variety of APIs, including open-source APIs like the Google Map API, and configured them in the primary application to support tourists in Bucharest by utilizing hardware API, which can make use of the location sensor and network interface in the tourists' smartphones to transfer data between the smartphones and the Google Map API. Nevertheless, the software might become useless and user data could be compromised due to the development team's poor development procedures ([Turing.com, 2021](#)).

Types of API:

Public API, also known as **Open API**, is one of the four principal categories of the Application Programming Interface. Others include **Private (Internal) API**, **Partner API**, and **Composite API** (Jacobson, et al., 2012).

Public or Open APIs are made available as open source by organizations for use by others in their application or software development process. In contrast, **Private or Internal APIs** are solely intended for use by those who work for the organization in question. Some businesses start out with private APIs but subsequently, make them available to the public with some limitations. The term "**Composite API**" refers to APIs that combine multiple data or service APIs and make one request to a server through numerous calls and receive one response with the necessary information. **Partner API** is an API that has been shared with partners or shareholders of an organization to enable them to access resources or increase revenue streams (IBM Corporation, 2012).

The **New York Times** API started as a private API and is transforming its business. "The NYT API grew out of a need to make our own internal content management system more accessible so that we could get the most from our content," said Derek Willis, Newsroom Developer at the *Times*. "The API offered a way to give more people access to create more interesting pieces. We are the biggest users of our own API, and that's not by accident. The API helps our business in other ways: creating brand awareness and helping us attract talent. But fundamentally, it helps us do our own jobs better (Jacobson, et al., 2012)."

API Technologies and Protocols:

Modern-day software and application development use various API technologies. **Some popular API Technologies are Web API, REST Architectural Style, SOAP, RPC, etc.**

Web API is a term used to describe an application programming interface (API) for a web server or web browser. It is a web development principle that frequently only pertains to the client side of online applications. This increases web applications' capabilities (IBM Corporation, 2012).

In a client-server architecture, **REST (Representational State Transfer)** is a kind of software architecture that defines a consistent interface between physically distinct components, frequently through the Internet. The creation, implementation, and

deployment of APIs may be done with greater freedom using the REST architecture, which separates the front and back ends of the API (IBM Corporation, 2012).

The high-level communications paradigm utilized by the operating system is provided by the **Remote Procedure Call (RPC)** protocol. **User Datagram Protocol (UDP)** or **Transmission Control Protocol/Internet Protocol (TCP/IP)** are two examples of low-level transport protocols that RPC makes use of to send message data between interacting applications. RPC develops a logical client-to-server communications framework to support network applications (IBM Corporation, 2012).

Simple Object Access Protocol (SOAP) is a messaging standard defined by the World Wide Web Consortium and its member editors. To enforce the structure of its payloads, SOAP employs an XML data format to specify both its request and response messages. SOAP serves as an interface for both public and private Application Programming Interfaces (APIs). Despite being more common in large businesses, SOAP APIs are created and used by companies of all sizes. Unlike other protocols and architectures, the SOAP protocol is independent of programming language, and **we can implement it in almost all modern programming languages like Python, Ruby, etc.** (Red Hat, Inc., 2022)

API Security:

API security protects **data** transferred through APIs, often between clients and servers linked over public networks. Organizations use APIs to connect services and convey data. Using **Transport Layer Security (TLS)** to implement data encryption and digital signatures is one of the many approaches to securing an API. Transport Layer Security safeguards data sent between you and a server and the privacy of your internet connection. As a means of authentication, **OAuth** is used in combination with **OpenID** as a technique to secure APIs. **OAuth**, which provides **authorization services**, may also be used to control who has access to which enterprise resources (API University, 2017).

To authenticate users and devices using OAuth 2.0, OpenID may use a third-party authentication system. This combination is currently one of the more secure authentication-authorization options on the market. If an API is breached, exposed, or hacked, it can disclose personal, financial, or other sensitive data, causing immense mayhem (Red Hat, Inc., 2019).

Conclusion:

To conclude the preceding arguments, while an API and its uses have numerous drawbacks if not adequately implemented, some of which were previously mentioned, an API is generally helpful because it saves resources and time for both developers and stakeholders who use applications or programs designed by linking various APIs together to make them functional and usable.

Through the help of an Application Programming Interface (API), we can use the functionality of one service, software or system with or within another one. For example, Google deploys its Google Maps API on its Find My Device software application to locate an authorized user's smartphone.

However, to benefit both the users and creators of these applications or programs, the developers must protect and authenticate the APIs they utilize to ensure that nothing unexpected or strange occurs.

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